

the main differences between Solid State Drives (SSDs) and Hard Disk Drives (HDDs):

Feature	Solid State Drive (SSD)	Hard Disk Drive (HDD)
Technology	Uses NAND flash memory chips to store data electronically.	Uses magnetic platters attached to a spinning motor and a mechanical read/write head.
Speed & Performance	Extremely fast. Boot times are significantly shorter, and files load almost instantly. Read/write speeds range from 500 MB/s to over 7,000 MB/s (for NVMe).	Slower. Bound by the physical limitations of the spinning disk. Read/write speeds typically range from 80 MB/s to 160 MB/s.
Durability	Highly durable. With no moving parts, they are highly resistant to physical shocks, drops, and vibrations.	Fragile. The mechanical read/write head and spinning platter can be easily damaged if dropped or jolted while running.
Lifespan	Limited by Terabytes Written (TBW) over time, but generally lasts many years under standard consumer use before memory cells degrade.	Not limited by read/write cycles, but eventually fails due to the physical wear and tear of the mechanical parts.
Noise & Power	Operates silently and consumes very little power, making them ideal for laptop battery life.	Produces a low humming or clicking noise. Requires more power to spin the disk and move the read/write arm.

Cost & Capacity	Higher cost per gigabyte. While larger capacities are becoming more affordable, they are still more expensive than HDDs of the same size.	Very cost-effective for large amounts of data. High-capacity drives (e.g., 4TB to 16TB+) are much cheaper than equivalent SSDs.
Typical Use Cases	Primary drive for operating systems, demanding applications (video editing, 3D rendering), gaming, and modern laptops.	Secondary drives for mass storage, archiving old files, network-attached storage (NAS), data backups, and budget desktop builds.